Ask My PDF Bot – Project Documentation

1. Introduction

The Ask My PDF Bot is an AI-based application that allows users to upload a PDF document and ask questions related to its content. The system reads the PDF, converts the text into embeddings using NLP models, stores them in a FAISS vector database, and retrieves relevant information to answer user queries using a Language Model (LLM).

2. Objective

The main objective of this project is to upload and process PDF documents, extract meaningful text, convert text into embeddings, store embeddings in FAISS, retrieve relevant chunks, and generate answers using an LLM.

Technology	Purpose
Python	Programming Language
Google Colab	Development Environment
PyPDF	PDF Text Extraction
Sentence Transformers	Embedding Generation
FAISS	Vector Similarity Search
Transformers	Language Model Pipeline
DistilGPT2	Answer Generation
NumPy	Numerical Computation

3. Technologies Used

Python, Google Colab, PyPDF, Sentence Transformers, FAISS, Transformers, DistilGPT2, and NumPy are used in this project.

4. System Architecture

Workflow: Upload PDF → Extract Text → Split into Chunks → Create Embeddings → Build FAISS Index → Ask Question → Retrieve Chunks → Generate Answer.

5. Working Principle

The system installs required libraries, imports modules, uploads PDF files, extracts text, splits text into chunks, creates embeddings using SentenceTransformer, stores embeddings in FAISS, retrieves relevant chunks based on user questions, and generates answers using DistilGPT2.

6. Retrieval-Augmented Generation (RAG)

The project follows the RAG architecture by combining a retrieval system (FAISS) with a generation model (LLM) to improve answer relevance and contextual understanding.

7. Advantages

Easy to use, fast document search, AI-powered question answering, lightweight implementation, and useful for beginners in Generative AI.

8. Limitations

Answers only one question, limited reasoning capability of GPT2, and increased processing time for large PDFs.

9. Future Enhancements

Future improvements include multi-question chatbot support, Streamlit web application, advanced LLM integration, chat history, and cloud deployment.

10. Applications

Useful in education systems, research document analysis, resume screening, legal document QA, medical report analysis, and enterprise knowledge systems.

11. Conclusion

This project demonstrates intelligent document understanding using NLP, FAISS vector search, and LLMs. It is a strong beginner-to-intermediate Generative AI project based on Retrieval-Augmented Generation (RAG).

12. Keywords

Artificial Intelligence, NLP, RAG, FAISS, LLM, Sentence Embeddings, Vector Database, PDF Question Answering, Transformers, Generative AI.